

Poster Design

How to get people to engage with your work, quickly!

Jeff Young, July 2025

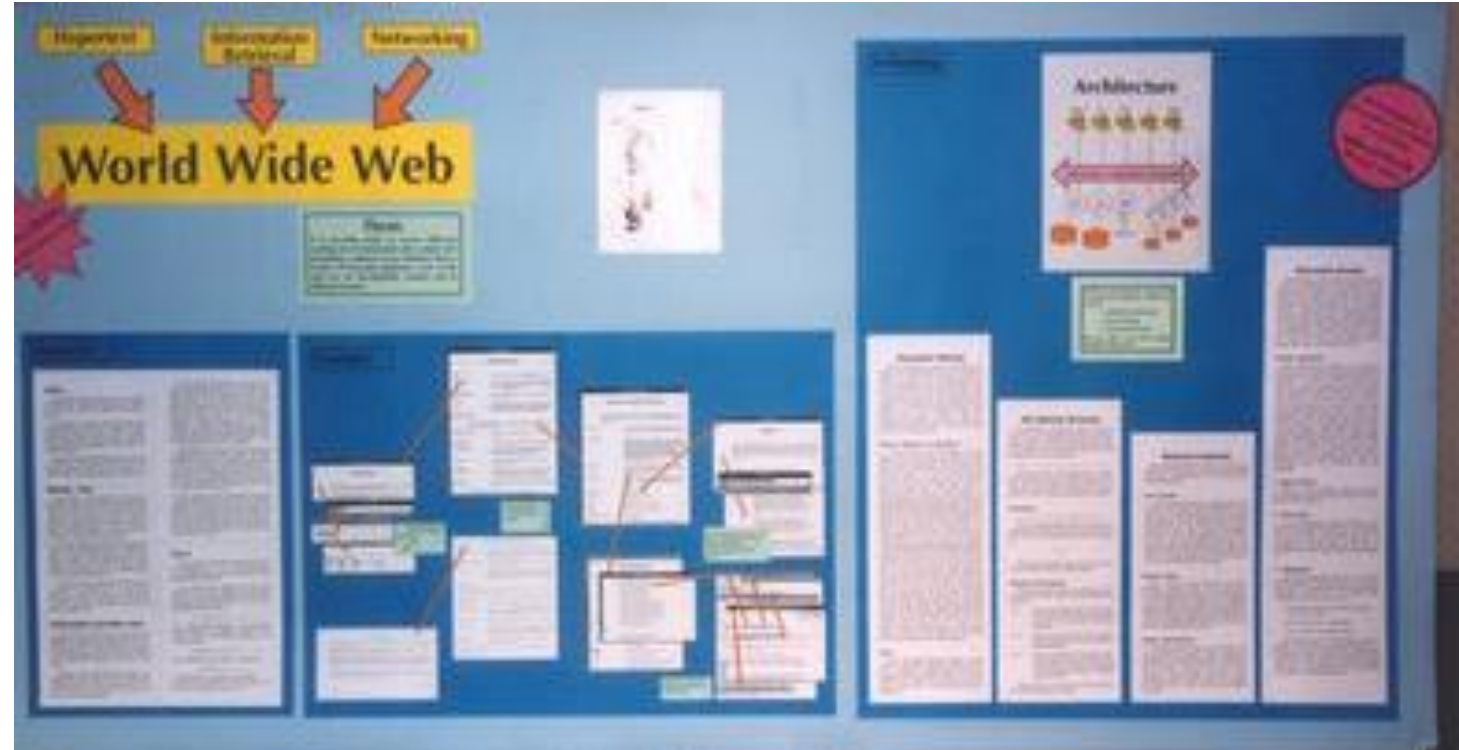
Thanks to Rich Vuduc for valuable feedback

Why Create a Poster?

Posters are sometimes seen as a “consolation prize” when a full-length talk or paper is not accepted

However, posters can successfully

- Communicate new and emerging concepts
- Quickly share key ideas that might go into a paper or presentation
- Use visual aids to engage viewers in a complicated topic



A blurry photo of Tim Berners-Lee's poster on the World Wide Web, 1991

What Makes a Good Poster?

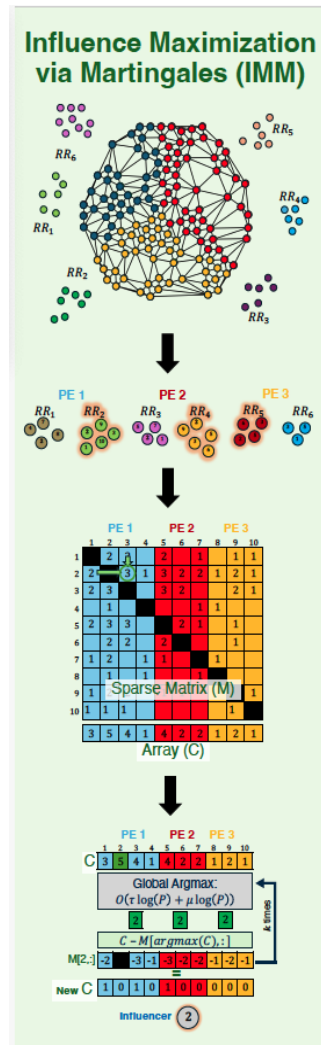
- Conveys an interesting idea or concept
- Demonstrates area of scientific impact

Is **easily parsed** at a high level

- The key point or contribution is clearly stated
- You want to avoid **“Tell me about your poster.”**

Backs up experimental studies or design with focused figures and results

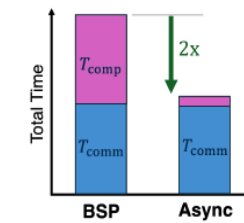
Allows the viewer to “follow up” and find more details on your work



Communicating more
can lead to faster
algorithms at scale

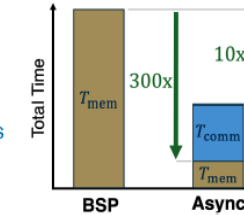
k-mer Counting

2x Speedup when we replace **non-blocking collectives** with aggregated **asynchronous point-to-point messages**



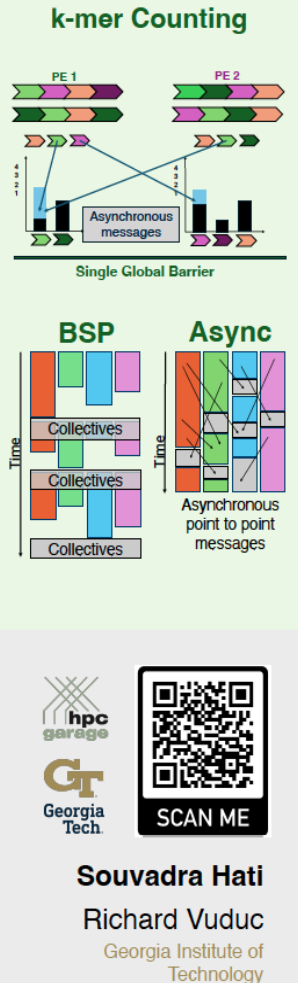
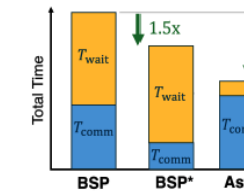
IMM Sampling

10x Speedup over communication avoiding BSP counterpart. **Copying** data results in **300x slower memory access**.



IMM Selection

45x Speedup using an algorithm performs **more communication** and uses **more memory** !! 🤔



Consider that quality poster design is really an experiment in user experience (UX)!

Key Elements

Sections – not all posters need all sections

- Title, Key Finding, and/or Takeaway
- Introduction of the problem being solved
- Tools/Methods – software being developed
- Results/Impact
- Citations/Acknowledgments
- Prompt to Action

Poster Layout

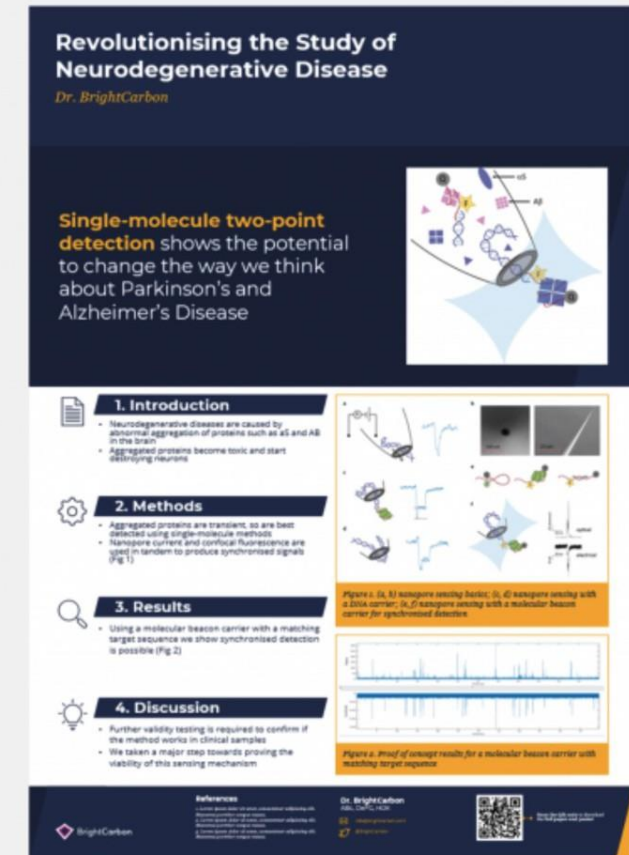
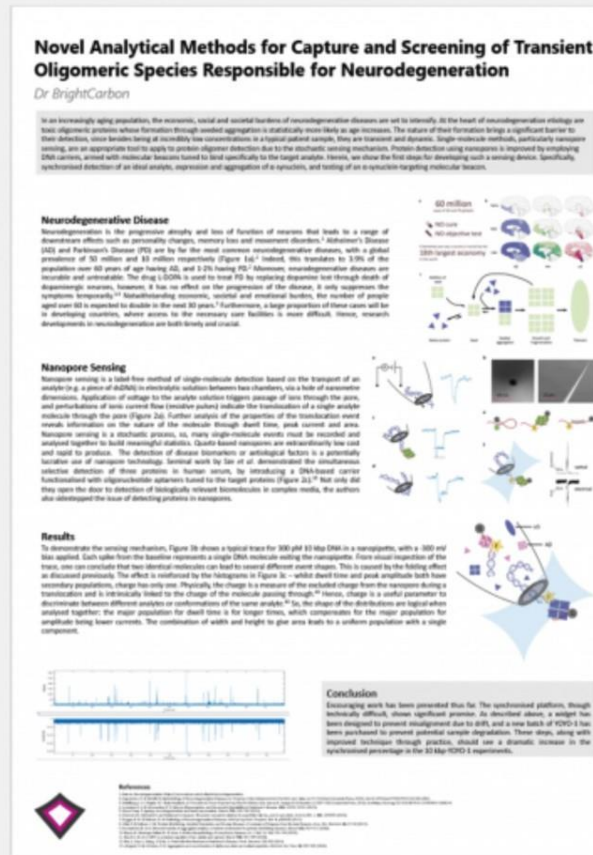
- Most posters use a two-column or three-column grid layout
- Gridlines or boxes can be used to effectively add white space
- Formatting is typically hierarchical

Remember: Your poster is an overview of the work you've done, ideally digestible in ~5 minutes by a reader

'I can hardly read this poster'



'Cool research, tell me more!'



<https://fourwaves.com/blog/how-to-make-a-scientific-poster/>

Dos and Don'ts

- **Do** make your poster easy to parse!
 - Simple, engaging titles where possible
 - Use human readable text
 - Make effective use of white space
 - Elaborate – include images/figures and simple text to reinforce the images
- **Do** include ways to further engage with the work
 - Authors, participants, citations
 - Links to repositories, QR codes
- **Don't** include “walls of text”
 - Most guidelines suggest using 500-1000 words
- **Don't** pack the poster with all your results or screenshots
 - Direct the viewer to additional resources; this is meant to be an overview!

What makes a great scientific poster...

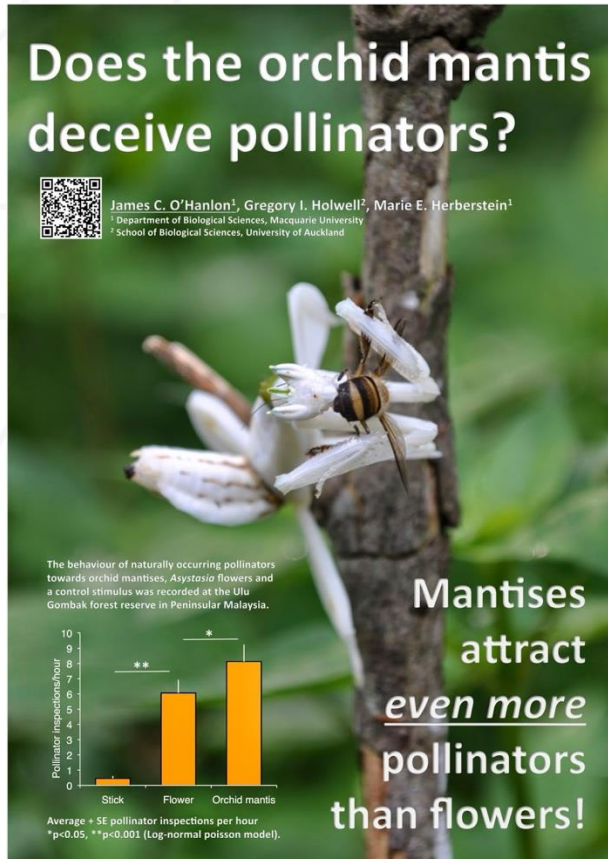


- | | |
|---|--|
| ✓ Important information can be read from 6-8 feet away. | ✗ Important information is hidden in dense paragraphs or is too small to read from several feet away. |
| ✓ The title is short, in plain language, and catches interest quickly. | ✗ The title is wordy or doesn't have any “hook” for passing viewers. |
| ✓ The word count is in the range of 300-800 (the less, the better). | ✗ The word count is more than 1000. |
| ✓ Bullet points, numbering, and emphasis on headlines makes for easy scanning and reading. | ✗ Content is organized into long paragraphs without headlines (or with headlines that don't stand out). |
| ✓ Graphics, colours, and fonts are visually appealing and help to communicate the message quickly. | ✗ No visual elements are incorporated (OR too many/poorly-thought-out visuals clash with each other and distract from the poster message). |
| ✓ The layout of the poster is consistent and clean with plenty of white space incorporated throughout the design. | ✗ The poster appears cramped and all available space is filled with info. |
| ✓ Acknowledgements, author name, and any institutional affiliations are clearly displayed on the poster. | ✗ No info on the author or institution is provided and it's unclear where the reader can go to find out more. |

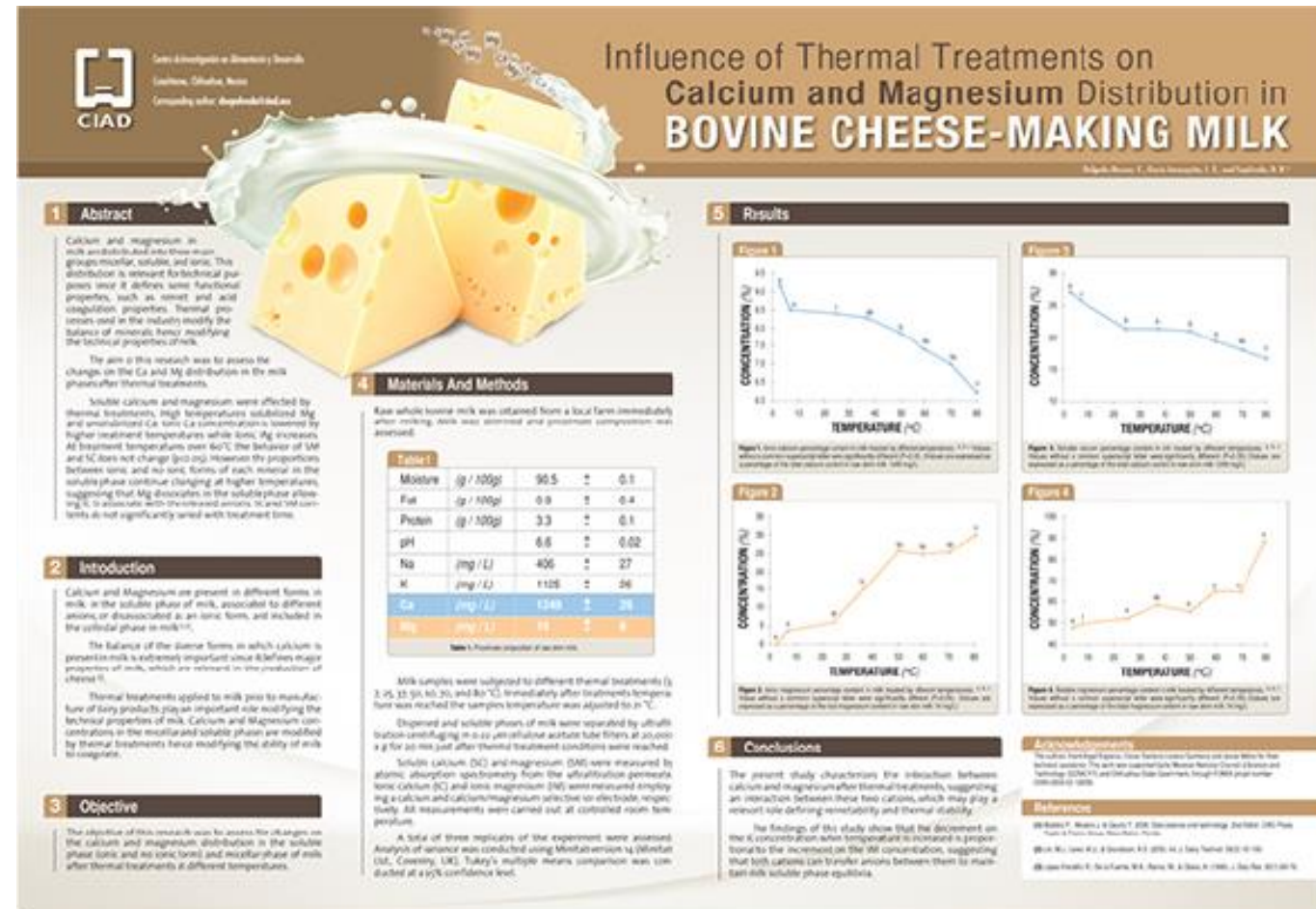


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Examples of Good Posters

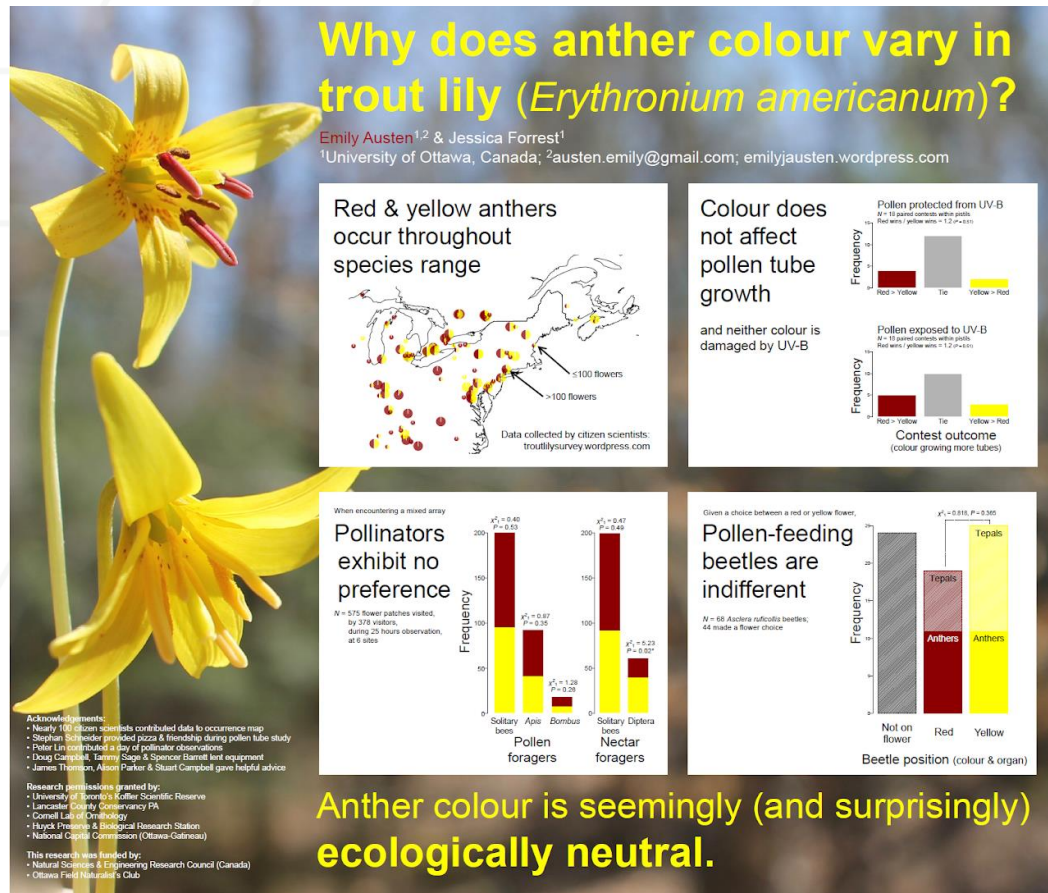


<https://web.archive.org/web/20160503015107/https://jamesohanlonresearch.wordpress.com/2015/01/09/conference-posters-less-is-more/>

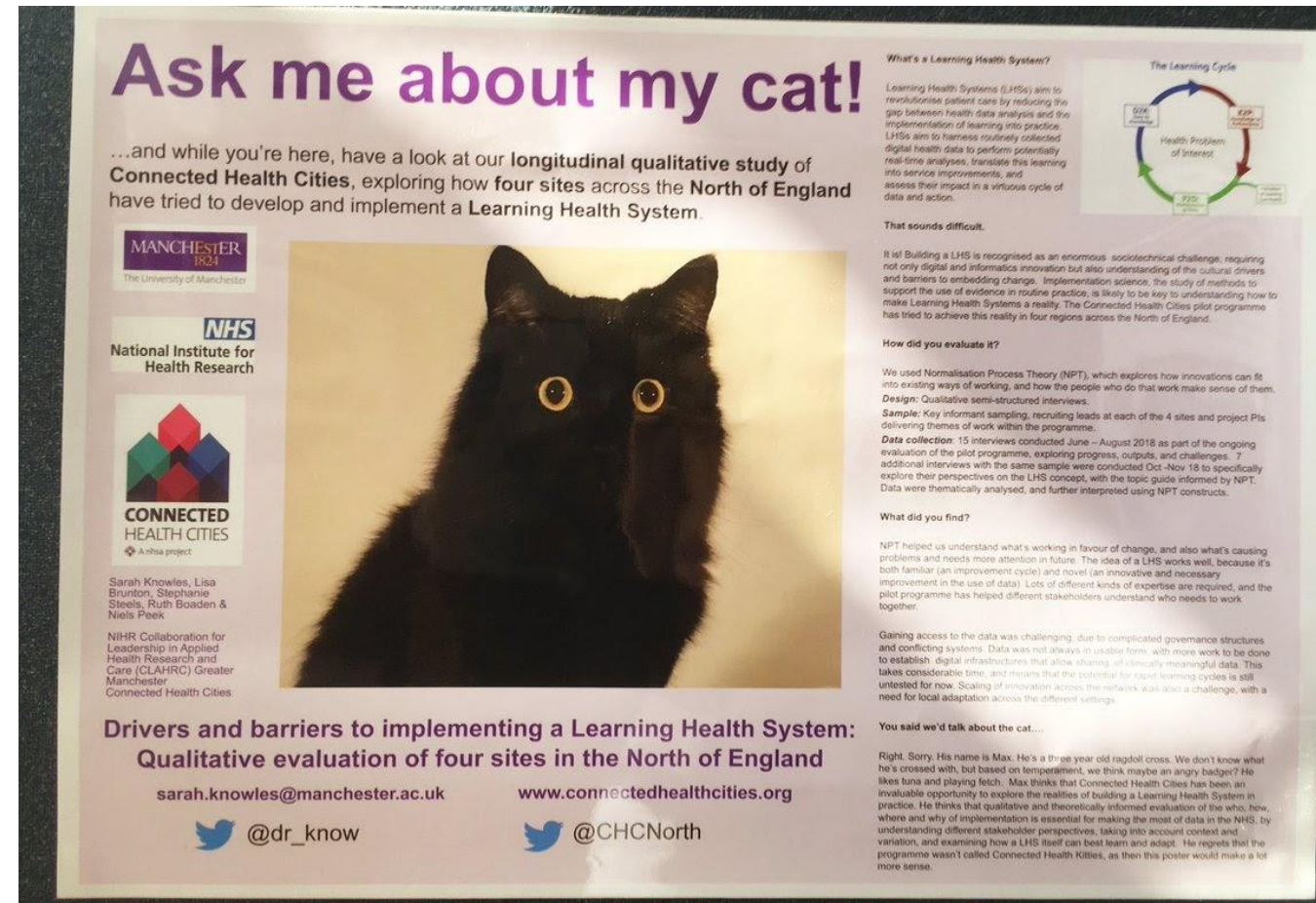


<https://www.animateyour.science/post/best-examples-of-scientific-posters>

Examples of Good Posters



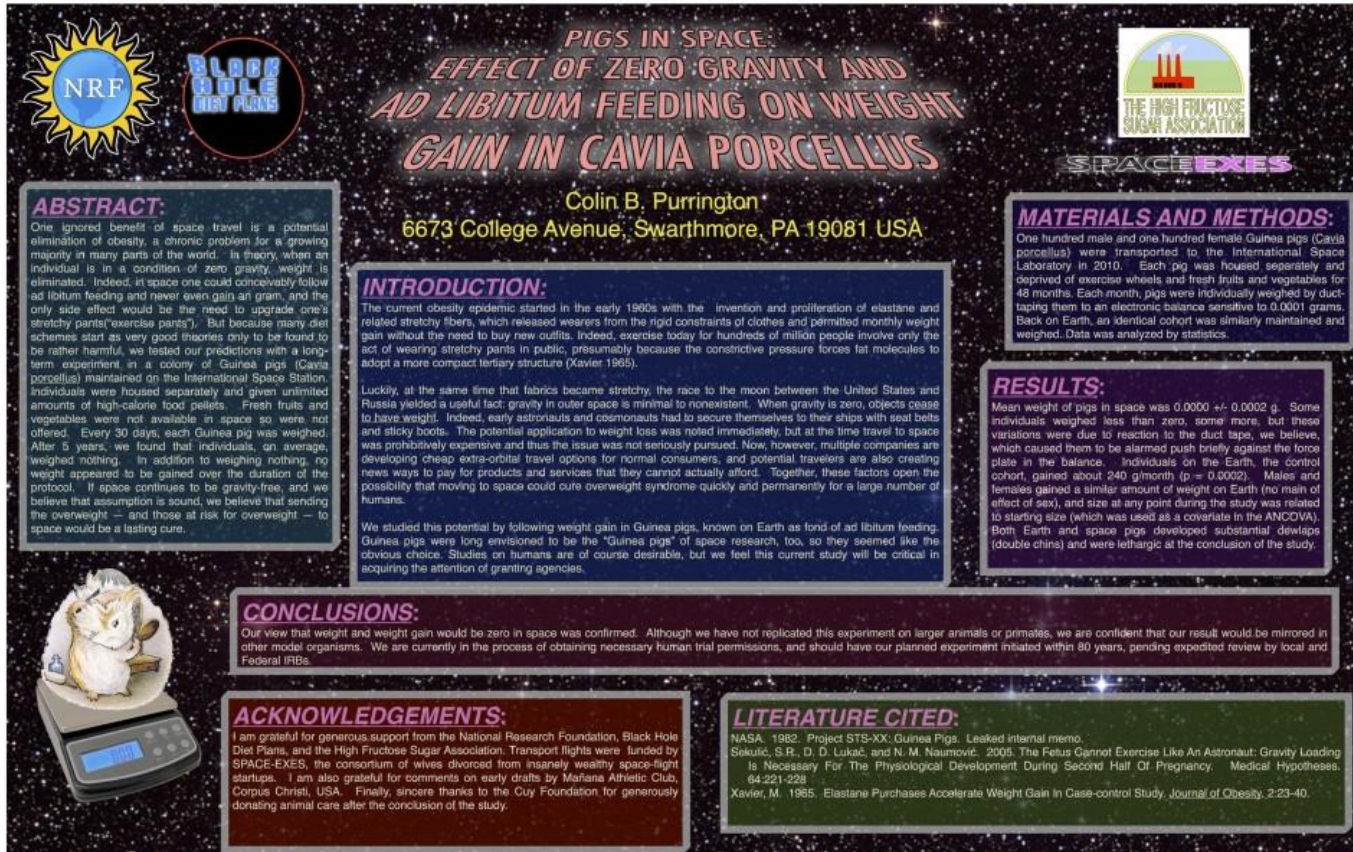
<https://betterposters.blogspot.com/2016/07/critique-anther-colours.html>



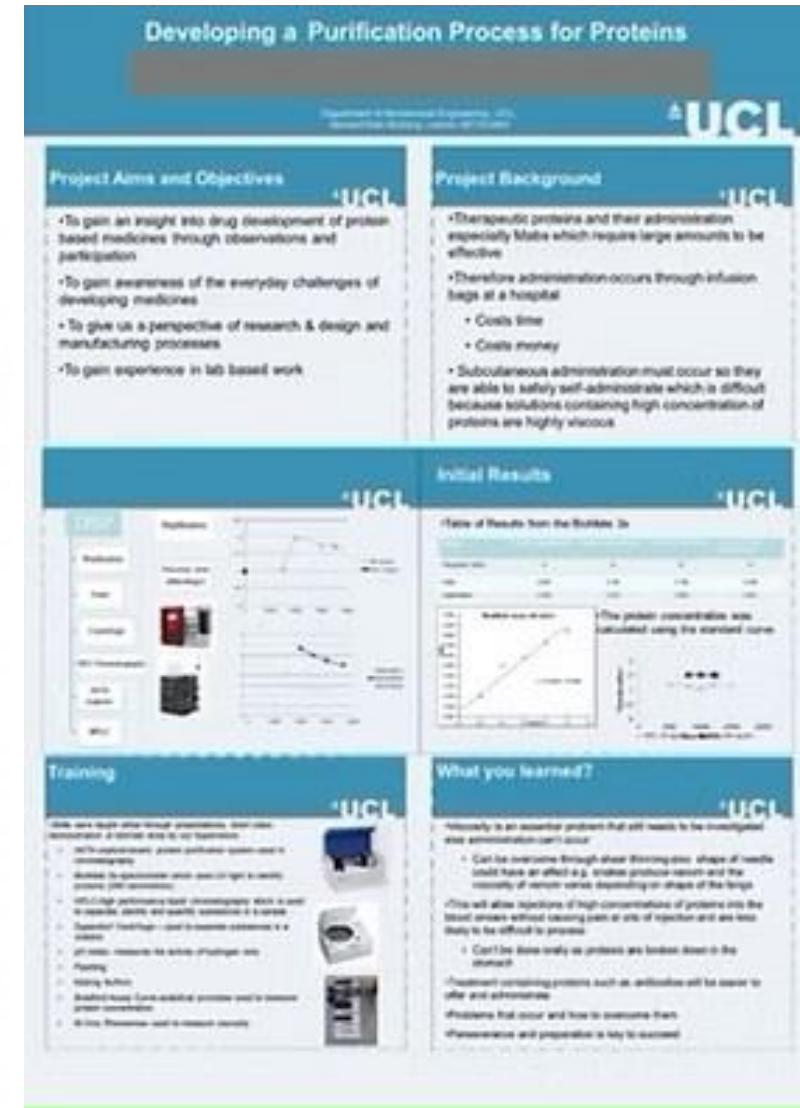
<https://sites.google.com/view/postergallery>

This is an example of “surprise and delight” as well as a useful “hook”. However, it doesn’t clearly show what the poster is about like the left example!

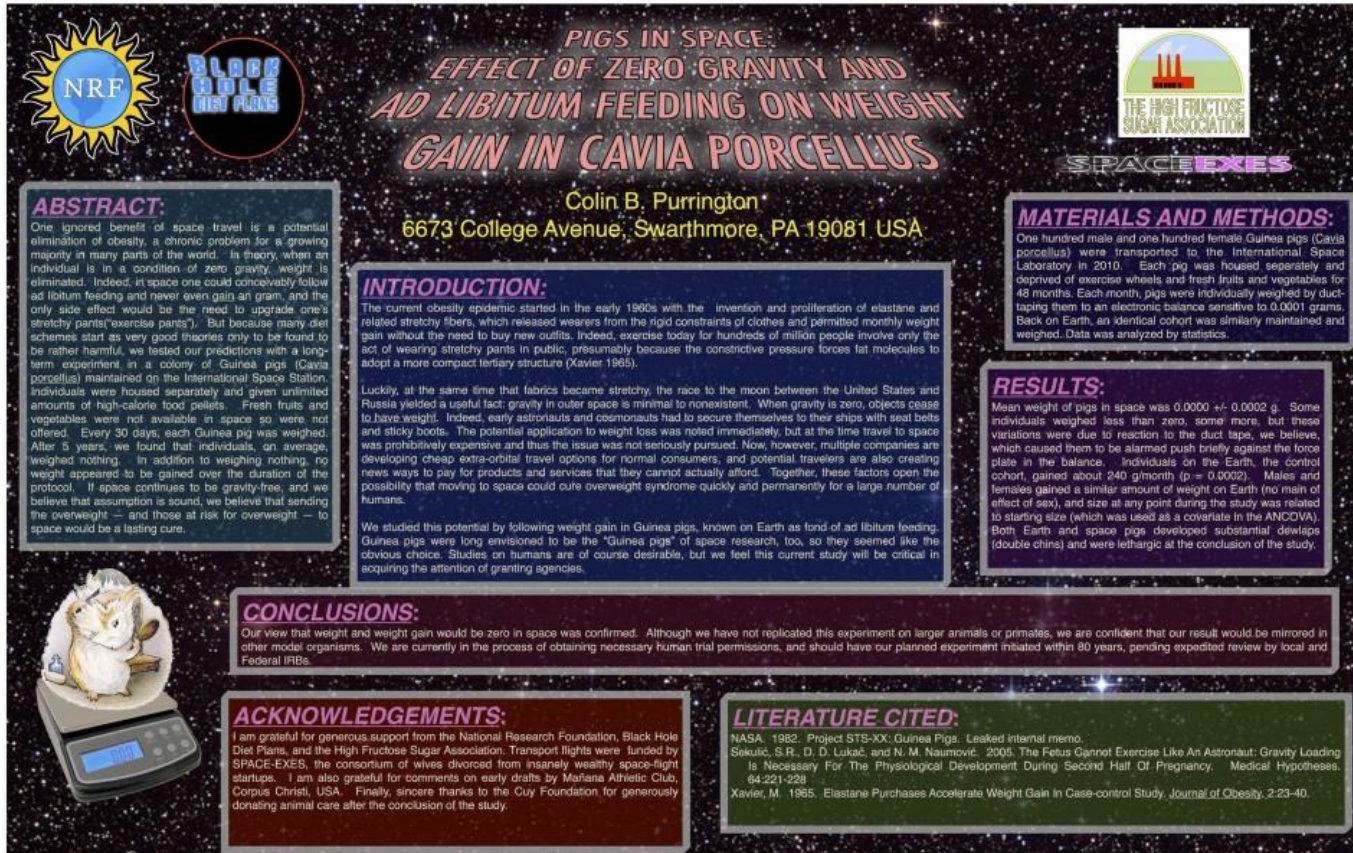
Examples of Bad Posters



copyright colin purrington
<http://colinpurrington.com/tips/academic/posterdesign>

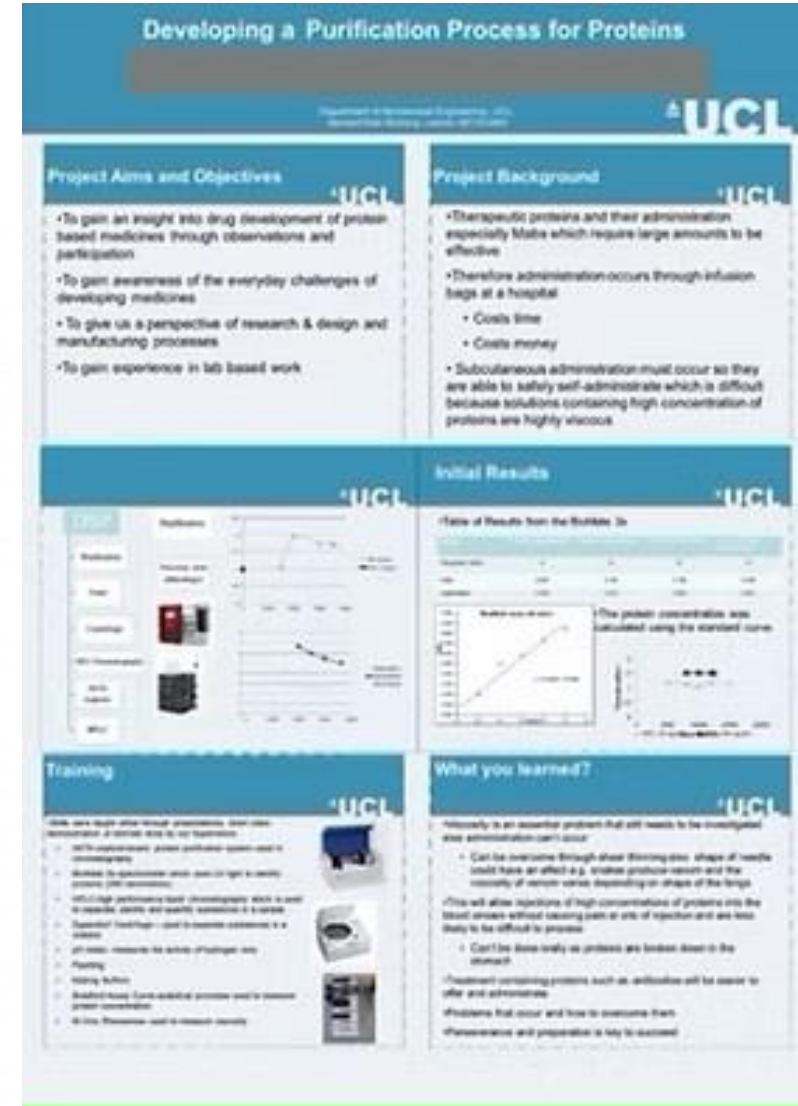


Examples of Bad Posters



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Walls of Text; Conflicting coloring, disorganization of boxes, no clear “hook” or problem statement



***Copied Slides
Directly to Poster!
Hard to Read***

What Do You See Here (Good or Bad)?

SSE Mission Statement

The Center for Scientific Software Engineering (CSSE) is focused on the development and dissemination of software engineering best practices to accelerate both the quality and pace of scientific discoveries at Georgia Tech and throughout the scientific community.

CSSE is located in the College of Computing and currently has a head of engineering and four full-time research software engineers.

What skills do our engineers have?

Software Engineering

Write code and apply best-of-industry practices to solve problems, automate processes and repetitive activities.

Software Architecture

Design software, infrastructure, and technology stacks that help propel a team forward.

Product Management

Apply lean software principles to validate hypotheses and develop minimum viable products (MVPs).

Program Management

Ensure successful outcomes by introducing practices that detect issues before they derail progress.

UI/UX Design

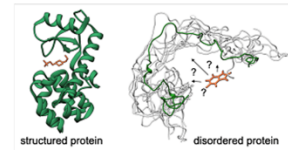
Create user interfaces and experiences that are efficient, engaging, easy to use, and require little to no training.

Technical Writing

Introduce prose that communicates complicated software ideas to broad audiences.

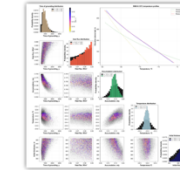
Highlights and Accomplishments

Targeting Intrinsically Disordered Proteins with Therapeutic Small Modules



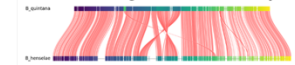
- Metainference simulations using CPU based molecular dynamics software
- Center engineers used profiling and testing to accelerate simulations
 - 53-68% improvement over CPU and GPU baselines

Numerical Heat-flow Model for Inferring Long-term Glacial History



- Jupyter notebook based code to model climate history from glacial samples
- Center engineers developed a Python package, CI/CD workflow, and PyPi deployment framework
- 20% improvement in performance

Fast Average Nucleotide Identity

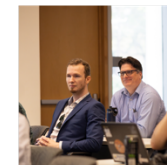


- Center partner in IDEaS refactored the widely used FastANI codebase used to find similarities in genomes
- Updated build system and added testing and CI/CD
- Newly released version of the code was widely adopted by the relevant community

Explore SSE projects at <https://gt-sse-center.github.io/project-showcase/>

Outreach to Campus and Beyond

CSSE supports workshops both at Georgia Tech and events at conferences like SIAM CSE, Supercomputing and US-RSE.



SSE engineers and faculty participate in the 2024 Software Sustainability Workshop

See our event resources at ssecenter.cc.gatech.edu

Collaborations with GT Units

Open Source Program Office Core Goals

Establish an OSS Standard for GT	Training	Sustainable OSS
Develop and share best practices for open-source software development at the institute level via an OSS research hub. How to create an open-source project. How to choose a license. How to develop an open source community.	Create and support a virtual summer internship experience targeted to introduce students to new open-source research projects and experiences. Technical Training & Workshops	Build the university's capacity to create, manage and maintain open-source software throughout the software lifecycle. e.g., Archival techniques to preserve open-source software

Outreach to Local Universities

CSSE interacts closely with our Open Source Program Office, an entity focused on supporting Georgia Tech open source projects and development.

Engaging with CSSE

There are several ways to work with SSE:

1. **Prototyping**
Create solutions that demonstrate system capabilities
2. **Partnering**
Augment a team's existing capabilities by working side-by-side for a predetermined set of time
3. **Guiding**
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6. **Profiling**
Scoped engagement to identify and propose resolutions to performance bottlenecks
7. **Curriculum Development and Training**
Develop and present curriculum that is broadly applicable to industry and academia

Learn more and contact us at ssecenter.cc.gatech.edu

Funding for CSSE is provided by Schmidt Sciences's VISS program

What Do You See Here (Good or Bad)?

Some possible answers:

Visually Clean

Effective Graphics

Good Follow-up Actions (could use a QR code)

May not have a clear “hook”

Poor Logical Flow (viewer may not be clear how to read/interact)

Logos are too high in the title

Unused white space on lower right

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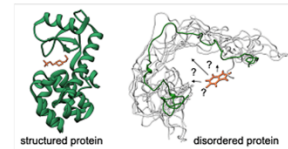
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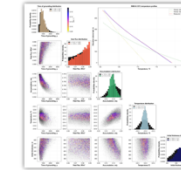
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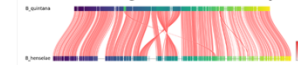
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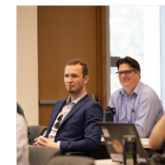


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Learn More – Poster Design Resources

Mike Morrison, “[How to create a better research poster in less time](#)”, YouTube

- Related Blog: <https://betterposters.blogspot.com/>

[Collin Purrington – Academic Poster Design](#)

[John Wilkes – Creating an Effective Poster](#)

[Matthieu Chartier, How to Make a Successful Scientific Poster](#)

[BioRender - Top 5 Tips for Designing a Scientific Poster](#)

[Miguel Balbin - Best Examples of Scientific Posters](#)